ChatGPT

NHS Digital-Therapy Procurement & Economics

Introduction

Digital mental health tools are playing an increasing role in NHS care, but moving from a small pilot to a paid NHS contract requires clear evidence of clinical benefit, economic value, and compliance with standards. OTOS – a digital ADHD and addiction recovery platform – aims to address the "Double Trouble Diagnosis" of co-occurring ADHD and substance use by delivering a hybrid of digital therapy and mentor coaching. To secure NHS commissioning within 12–18 months (after an initial Cambridgeshire & Peterborough pilot), OTOS must demonstrate robust outcomes (e.g. reduced relapse, improved medication adherence) and cost savings, while aligning with NHS evaluation frameworks and procurement pathways. Below, we outline how digital mental-health tools transition from pilot to commissioned services in England (with notes on Scotland/Wales/NI where relevant), focusing on economic models, evidence frameworks (NICE's Evidence Standards, DTAC, NIHR), procurement routes (accelerators, AHSNs, ICS purchasing), and real examples like Sleepio, SilverCloud, and Wysa. This analysis culminates in actionable guidance for OTOS to build a pilot proposal that is "commissioning-ready."

Evaluation Frameworks & Standards

NICE Evidence Standards Framework (ESF): NHS commissioners expect digital health innovations to meet the NICE ESF for Digital Health Technologies. This framework classifies digital tools by function/risk and defines the level of evidence needed to demonstrate effectiveness and value [1] [2]. For a solution like OTOS (which offers therapeutic intervention for diagnosed conditions), a higher evidence tier is likely – meaning robust clinical evidence (at minimum real-world pilot outcomes, and ideally comparative or controlled data). Meeting the ESF doesn't equate to formal NICE endorsement, but it **guides NHS evaluators on whether a product is ready for adoption** [3] [4]. OTOS's pilot should be designed to generate evidence aligned with the ESF: for example, use validated clinical outcome measures (ADHD symptom scales, relapse rates, anxiety/depression scales) and even a control or comparator if possible. The goal is to show significant improvement in patient outcomes with OTOS versus standard care. High-quality evidence will position OTOS well for later NICE appraisal or inclusion in NHS pathways. Notably, **NICE and NHS England have provided tools like an ESF user guide and a budget-impact spreadsheet** to help innovators plan evaluations [5] – OTOS can leverage these to estimate its potential cost-benefit in NHS context.

Digital Technology Assessment Criteria (DTAC): In parallel with clinical evidence, NHS purchasers will require that OTOS meets the **Digital Technology Assessment Criteria**, the national baseline for digital health tool compliance. The DTAC covers five key domains: *clinical safety, data protection, technical security, interoperability, and usability/accessibility* [6] [7]. All new digital technologies – even during pilot trials – should be assessed against DTAC to ensure minimum standards are met [8]. Practically, this means OTOS must have: a clinical safety officer and **DCB0129-compliant safety case** (demonstrating risk management of the software in a healthcare setting); robust data privacy and GDPR compliance; cybersecurity measures; integration capabilities (e.g. ability to share data with NHS systems like electronic patient records); and evidence of user-centered design (accessible interface, tested with users including those with disabilities). **Wysa's experience is instructive**: Wysa became the first AI mental health app to meet the NHS DCB0129 clinical safety standard [9], and being **"DTAC-ready" is now**

expected at procurement [10] [7]. OTOS should engage early with NHS digital governance teams (or use third-party assessors) to **prepare a DTAC file** documenting compliance in each area. This will give NHS stakeholders confidence that the platform is safe and can be deployed within NHS IT environments. Achieving an ORCHA accreditation or listing on the NHS Apps Library can further validate that OTOS is a **credible, trusted tool** – Wysa, for example, is top-rated by ORCHA, which helped build trust with NHS providers [9]. In summary, **strong clinical evidence and DTAC compliance go hand-in-hand**: NHS commissioners want proof that a digital therapy both *works* and is *safe/secure* before they will fund it [1].

NIHR and Real-World Evaluation: To move from a pilot to a business-as-usual service, OTOS will benefit from independent or academic evaluation of its impact. The National Institute for Health and Care Research (NIHR) often supports studies of digital health innovations in NHS settings, ensuring rigorous outcome and economic analysis. Incorporating NIHR-aligned evaluation methods in the pilot (even if the pilot isn't formally NIHR-funded) will strengthen the case for commissioning. For instance, OTOS should define outcomes like reduced relapse rates, treatment retention, or quality-of-life improvements, and measure these with validated instruments. It should also collect resource-use data (e.g. number of GP visits, inpatient days, or care coordinator contacts for participants vs. baseline) to feed into a health-economic analysis (using, say, the EQ-5D for quality-adjusted life years and a cost offset analysis) [11] [12]. NIHR expects reporting on both clinical outcomes and cost-effectiveness; OTOS's pilot could partner with an academic group (perhaps via the local NIHR Applied Research Collaboration in East of England) to design the study and analyze results. The **NHS Innovation Service** (an NIHR-supported hub) can also connect innovators to evaluation partners [13]. By planning for publications or NIHR reports, OTOS can ensure that at 12–18 months, it has credible data on *what the intervention achieved* and *how that translates to NHS value*. This will be vital for convincing Integrated Care Systems (ICSs) and budget-holders.

Economic Models & Cost Savings

A compelling economic model is often the linchpin in converting a pilot into a paid contract. NHS commissioners will ask: **"Does this digital therapy save us money or resources, compared to current care?"** OTOS should therefore emerge from its pilot with a clear **cost-per-patient figure and a return on investment (ROI)** analysis. Here's how to approach it:

- **Cost-Per-Patient Calculation:** Determine the all-in cost to deliver OTOS per patient (for a standard program length, e.g. 12 weeks). This includes the platform usage cost, any licensing or subscription fee, and the cost of the mentor/coach time per user. For instance, if one OTOS mentor can support X patients concurrently, allocate their salary cost appropriately. This per-patient cost will be the basis for pricing discussions with the NHS (e.g. a yearly license per user or per population). Digital therapies often benefit from economies of scale – serving more users marginally increases costs far less than traditional therapy – so highlight that scalability. **Benchmark against similar tools:** Big Health's Sleepio (digital CBT for insomnia) was evaluated at roughly **£55–65 per patient for a full course** in early implementations, and that was shown to be cost-saving within one year [14]. OTOS might similarly target a per-patient cost in the low hundreds of pounds for a multi-week program.

- **Comparative Value (Cost Offset):** To show ROI, compare OTOS's cost against the costs of standard care or negative outcomes that it can prevent. For example, relapse of substance use or unmanaged ADHD can lead to expensive outcomes: inpatient detox or rehab stays, A&E visits, additional mental health appointments, or even justice system costs. If OTOS can reduce relapse rates or symptom severity, those avoided costs are part of the ROI. A vivid comparison can be made with rehab: **one residential rehab bed for a month can cost around £2,400–£4,000** (often ~£600 per week in local authority-funded rehabs) [15] [16]. For that same expenditure, an NHS commissioner could fund on the order of *a dozen patients* through a 12-week OTOS digital recovery program. In other words, *the cost of treating one person in inpatient rehab for 4 weeks could cover 12 people using OTOS for 3 months*. This kind of comparison grabs attention – it underscores how digital solutions can expand access at a fraction of the cost. OTOS should back up such comparisons with pilot data (e.g. "in our pilot, 12 weeks of OTOS support cost £X per patient, versus £Y for traditional care").

- **ROI to NHS/ICS:** The pilot should track healthcare utilization for participants to quantify "cashable" savings. For instance, do OTOS users have fewer unplanned GP visits, A&E attendances, or psychiatric admissions compared to a similar cohort not using OTOS? If OTOS supports medication adherence, perhaps fewer GP appointments are needed for managing ADHD medications or co-morbid conditions. **Sleepio's evaluations provide a model**: in the Thames Valley pilot, over 1,200 patients used Sleepio, which led to significant reductions in GP consultations and sleep medication prescribing – yielding a projected **£106 saved per patient** entering CBT treatment and an overall £1.8 million saving when scaled to the whole ICS population [17]. Likewise, **NICE's guidance on Sleepio confirmed it as an in-year cost-saving** measure for the NHS, primarily due to reduced GP visits and less pharmacotherapy for insomnia [14]. OTOS should aim to produce a similar calculus: e.g. "OTOS users had X% fewer A&E visits over 6 months, saving £ ***per patient in acute care costs," or "OTOS helped Y% of patients maintain abstinence, potentially avoiding £*** in future inpatient admissions." Even if the pilot is small, **extrapolate the impact** to illustrate system-level savings if scaled (as the Office of Health Economics did for Sleepio, projecting ~£99 million national savings over three years based on pilot outcomes [18]). Another example: the Wysa chatbot, used in NHS Talking Therapies, costs only **~£5.90 per user per year**, and by handling initial triage and self-management, it can reduce the load of patients requiring full therapist-led care; with ~440k referrals a year not receiving face-to-face help, the time saved in triage (97 minutes per referral at £105 cost each) can translate to substantial savings when such AI support is in place [19]. OTOS can similarly argue that its hybrid support will lower overall care costs by pre-empting crises and augmenting understaffed services.

- **Return on Investment Narrative:** Present the ROI in a way that resonates with ICS priorities. For example: "For every £1 spent on OTOS, the NHS gets £3 back in reduced treatment costs or productivity gains within 1 year," if pilot data can support that. Even qualitative ROI can help – **SilverCloud's digital CBT, when integrated into NHS IAPT services, was shown to be effective and *potentially cost-effective in the long term**, with up to 91% probability of cost-effectiveness at standard NHS thresholds by 12 months [12]. This suggests that scaling digital self-guided therapy yields a net positive economic benefit to the health system. Similarly, investment in mental health has a broader economic return (estimates of 3.3–5.7 to 1 when including societal benefits) [20] – commissioners are aware of these paradigms. OTOS should emphasize both direct NHS savings and indirect benefits (for instance, improved functioning of individuals with ADHD may lead to better employment or less social care usage, which aligns with ICS outcomes frameworks).

In summary, **the pilot's business case must translate clinical outcomes into £**. OTOS should produce a draft economic model from pilot results – possibly using NICE's budgeting tools [21] – to forecast how an ICS commissioning OTOS could reinvest savings elsewhere. Numbers backed by real data (even if from a small sample) will carry weight in funding decisions.

Procurement and Scaling Pathways

Having strong evidence and a favorable economic case sets the stage, but OTOS will also need to navigate the NHS innovation and procurement landscape to secure contracts. Key routes and mechanisms include:

- **Local/Regional NHS Commissioning (ICS route):** In England, Integrated Care Systems control budgets and decide on service innovations. A typical pathway is: **pilot within one part of the system → demonstrate value → make the case for wider ICS adoption**. For example, OTOS's initial pilot with Cambridgeshire & Peterborough NHS (through partners like CGL/Change Grow Live, CRS, or CPFT) can act as a proof-of-concept. Success there should be leveraged to approach the ICS digital transformation board or the Mental Health programme leads to fund a rollout across the ICS. Often, if a pilot is grant-funded (say by an innovation fund or AHSN) and shows good outcomes, the ICS may pick up the ongoing costs afterward. It's crucial to identify **which budget pot** would fund OTOS in steady state: since OTOS is positioned as a mental health innovation (supporting ADHD and addictive behaviors), funding would likely come from the mental health or community services budget of the ICS, rather than siloed substance misuse funding (which is usually local-authority commissioned). Framing OTOS under **"integrated mental health care"** aligns it with NHS priorities (e.g. improving community mental health, expanding IAPT/Talking Therapies, tackling health inequalities) and avoids the scenario of falling between commissioners. As an example, a digital mental health tool for insomnia might be picked up by an ICS's mental health programme rather than expecting GP prescribing budgets to pay – after Sleepio's pilot success, some regions (like the **Buckinghamshire, Oxfordshire, and Berkshire West ICS**) made Sleepio available to their populations, citing adherence to NICE guidance and cost-effectiveness in "challenging financial conditions" [17]. Similarly, OTOS should align its value with ICS mental health objectives (e.g. reduce unmet need in adult ADHD, support "dual diagnosis" patients who often bounce between services).

- **NHS Innovation Accelerator (NIA) and AHSN Support:** Many digital health tools have accelerated their NHS adoption by joining formal innovation programs. The **NHS Innovation Accelerator (NIA)** is a national program that selects a cohort of promising innovations each year and helps them scale through mentorship and networking across the NHS [22] [23]. Wysa, for instance, was accepted into the NIA in 2023, which gave it a platform to reach multiple NHS trusts and ICSs simultaneously [24]. Being an NIA Fellow comes with endorsement that "this innovation is NHS-approved for scaling," which can reassure local commissioners. OTOS should consider applying to the NIA once some pilot evidence is in hand (the timing could align with the next NIA intake). In addition, Academic Health Science Networks (AHSNs) at the regional level can champion new solutions. Eastern AHSN (covering Cambridgeshire/East of England) is particularly relevant – they have a track record of supporting digital mental health pilots (Oxford AHSN's backing was pivotal for Sleepio's first large rollout and evaluation [25]). Engaging the AHSN early can help in **designing the evaluation**, identifying NHS partners, and later spreading OTOS to other ICSs in the region. The AHSN might also connect OTOS to **NIHR or SBRI Healthcare funding** opportunities (Small Business Research Initiative competitions) to fund further trials or development. For example, Big Health secured Innovate UK funding through such channels to implement Sleepio at scale and gather evidence [26] [27] – those data later underpinning the case to NICE and the NHS. OTOS could similarly seek an innovation grant to subsidize the pilot, making it low-risk for the NHS site, in exchange for robust data sharing.

- **HealthTech Hubs and Digital Accelerators:** Aside from the NIA, there are regional accelerators (often run by AHSNs or partners). For instance, **DigitalHealth.London Accelerator** has supported companies like SilverCloud and Ieso to integrate into NHS pathways. Wysa participated in the *DigitalHealth.London* program and even won an award for NHS impact [28]. These programs provide networking with commissioners, sharpen the product's fit for NHS needs (e.g. integration into care pathways), and can generate publicity. In the East of England, OTOS can tap into networks like the East of England Innovation Hub or the **NIHR East of England ARC** for implementation science support. There are also initiatives like the NHS **Mental Health Tech Innovation hubs** (some regions have clusters focusing on mental health digital tools). OTOS should map out relevant hubs – for example, Greater Manchester has a strong digital health scene via Health Innovation Manchester, and Yorkshire & Humber AHSN has an Propel Digital accelerator, etc. Participation can ease the pathway into those ICSs by demonstrating local commitment to innovation.

- **Procurement Frameworks:** When an ICS or NHS provider is ready to buy OTOS, using established procurement frameworks can simplify the contracting. OTOS should ensure it is available through routes like the **NHS Supply Chain or G-Cloud (UK Digital Marketplace)**. For example, SilverCloud and other digital therapies often list their services on the NHS Digital Marketplace (G-Cloud/DPS) [29], which allows NHS orgs to procure without a full tender by calling off an approved contract. Similarly, being on framework agreements (like "Health Systems Support Framework" or relevant NHS England procurement frameworks for digital interventions) can remove barriers to purchase. While these details are more about paperwork, it's *actionable* for OTOS to engage with a procurement advisor or use the new NHS **"Apply to Supply" portal** to get on approved supplier lists [29]. Essentially, **make it as easy as possible for an NHS manager to sign on the dotted line** – all compliance checked (via DTAC), data security cleared (Info Governance), and a contracting vehicle in place.

- **NHS England or Devolved-Nation Initiatives:** In some cases, national or devolved-nation health bodies will directly procure digital tools after strong evidence is established. Scotland provides a compelling example: the Scottish Government, after running initial pilots, struck a **nationwide deal with Big Health** to offer Sleepio (for insomnia) and Daylight (for anxiety) across all NHS Scotland boards – touting it as a "world-first" national digital therapeutic rollout [30] [31]. This was predicated on proven outcomes (70% of users improved and gained ~7 hours sleep/week in earlier deployments) and the context of COVID-19 increasing demand for mental health support [31]. While OTOS's first goal is local adoption in England, it's worth noting that **Wales and Northern Ireland** often observe NICE guidance and may adopt digital therapies once England does. NHS Wales has its own digital health ecosystem (including Health Technology Wales, which uses the NICE evidence framework to decide on appraisals [32]). If OTOS can pioneer an evidence base in England, a next step could be to approach NHS Wales or NHS Scotland with that data for a broader implementation – especially since the *postcode lottery* of digital mental health was criticized after Scotland's deal [33]. In England, NHS England itself has run programs like the **IAPT digital therapy pilots** (several years ago, NHS England evaluated online CBT tools centrally). If similar central initiatives arise (for example, an NHS England fund for digital mental health to cut waiting lists), OTOS should be ready to apply. Keeping an eye on NHS Long Term Plan funding streams (e.g. for community mental health or health inequalities) can uncover opportunities to get central backing.

In short, **the strategy is:** prove locally, then scale regionally/nationally. Use the **"best practice" channels – AHSNs, NIA, accelerators – to gain endorsements and spread awareness.** This network effect can turn a successful 30-patient pilot in Cambridge into interest from Manchester, Liverpool, or Birmingham where ADHD and addiction comorbidities are pressing issues. Indeed, regions like Merseyside, Greater Manchester, and the West Midlands have highlighted needs in dual diagnosis and often have innovation funds targeting mental health. OTOS can target those ICSs for second-wave pilots or adoption, referencing the outcomes from Cambridgeshire. Having champions helps: identify a clinical champion (e.g. an NHS psychiatrist or GP in the pilot) who can present the results to peer networks and at conferences.

Case Studies: Pilots to Commissioned Services

Real-world examples of digital mental health tools offer valuable lessons for OTOS:

Sleepio (Digital CBT for Insomnia)

Sleepio by Big Health is a standout case of a pilot translating into wider NHS commissioning. **Initial pilots:** Supported by Innovate UK and Oxford AHSN, Sleepio was rolled out to ~28,000 people in the Thames Valley region via self-referral and GP/IAPT referrals [34] [35]. Outcomes from the pilot were impressive – a **56% reduction in use of sleeping pills and a 70% anxiety symptom reduction** among users [18]. These clinical gains were accompanied by economic analysis: the Office of Health Economics evaluated Sleepio's impact across 9 GP practices, finding significant cost savings. The projection was **£130k saved over 15 months in that sample, equating to ~£1.8 million across the whole ICS over 3 years (≈£106 saved per patient)** [17]. On the strength of this evidence, Sleepio became the **first digital therapeutic to receive a positive NICE guidance** – in 2022 NICE recommended Sleepio as a safe, clinically effective, and *cost-saving* option for insomnia [14]. The NICE committee noted Sleepio's ability to reduce GP visits and medication costs by giving instant CBT access where therapists were unavailable [14]. This endorsement was pivotal: it "gives NHS commissioners and clinicians added confidence in funding and prescribing Sleepio" [36]. Following NICE approval, Scotland adopted Sleepio nationally (free to all adults via self-referral) as part of its digital mental health strategy [30] [31]. In England, while NHS England did not immediately fund a nationwide rollout, multiple ICSs and NHS trusts proceeded to contract Sleepio, leveraging the NICE guidance and local business cases. **Lessons for OTOS:** a strong partnership with an AHSN and use of independent health-economic evaluation can create a powerful value story. Furthermore, achieving NICE guidance (even if time-consuming) is a "gold stamp" that can greatly accelerate commissioning decisions. Sleepio's journey also shows the importance of addressing a known NHS gap (insomnia therapy) and collecting data on both clinical and utilization outcomes.

SilverCloud (Online CBT Platform)

SilverCloud Health provides a digital platform for guided self-help CBT, and it has been integrated into NHS services at scale. SilverCloud's path started with research trials and small pilots in IAPT (Talking Therapies) services. Early **randomized controlled trials** demonstrated that SilverCloud's online modules for depression/anxiety, when used with brief supporter guidance, were effective at reducing PHQ-9 and GAD-7 scores and could be delivered at lower cost than traditional therapy. A 2020 pragmatic trial within NHS IAPT showed significant improvements in depression and anxiety with iCBT, and when outcomes were modeled over 12 months, the probability of cost-effectiveness exceeded 90% at standard NHS thresholds [12]. With this growing evidence base, NHS England ran an initiative to evaluate digital therapy options, and SilverCloud was one of the approved solutions made available to local services. Adoption then accelerated: many NHS Trusts and ICSs commissioned SilverCloud to expand access and help meet waiting-time targets. By integrating with the stepped-care model, SilverCloud allowed patients with mild-moderate symptoms to start therapy online immediately, reserving human therapist time for more severe cases. Over time, SilverCloud reported reaching over 300,000 NHS users [37] [38]. Its outcomes were monitored, and data showed that **12-month post-treatment benefits were sustained and the addition of digital CBT improved recovery rates cost-effectively** [39] [40]. SilverCloud's success was also enabled by partnerships – e.g. an NIHR-funded project "Digital IAPT" helped evaluate and refine its use in routine care. **Lessons for OTOS:** integrating into existing care pathways (like IAPT or ADHD clinics) can facilitate uptake – you become a tool that existing staff can use to treat more patients. Having solid clinical trial data lends credibility, but real-world service data (on engagement levels, outcome improvement, etc.) is equally important for NHS decision-makers. SilverCloud also showed the value of persistence: over nearly a decade it went from small pilots to a mainstream solution, in part by aligning with NHS England's strategic push for digital and by continuously gathering evidence of long-term outcomes and cost-effectiveness [12].

Wysa (AI Mental Health Chatbot)

Wysa offers an AI-driven conversational agent that supports mental health self-management and augments therapy. Its journey reflects how a novel technology can rapidly scale in the NHS when it addresses capacity problems. Wysa began as a direct-to-consumer app, but by the late 2010s it engaged with NHS pilot programs – for example, offering the app to NHS staff during the COVID-19 pandemic, and pilot use in certain Talking Therapies services for waitlist support. **Crucially, Wysa invested in meeting NHS standards early:** it worked closely with regulatory bodies, becoming the first AI mental health app to complete the DCB0129 clinical safety certification [9]. It also achieved ORCHA approval and won privacy awards, easing NHS concerns about risk [9]. As a result, Wysa gained traction; by 2022 it reportedly had contracts with more than a dozen NHS trusts, making it one of the "biggest suppliers of AI to the NHS" for mental health [41]. Wysa's value proposition to NHS services is to *save clinician time* and *improve access*: It can triage patients via an interactive chat instead of static forms, provide guided self-help exercises to those on waiting lists, and even deliver a form of CBT for those who might otherwise drop out [42] [43]. An independent real-world study found Wysa engagement led to improvements in depression and anxiety scores, validating its efficacy as an adjunct support [44]. Economically, Wysa's model is compelling: at just **£5.90 per eligible user per year**, it can potentially reduce thousands of unnecessary clinical triage sessions, which cost over £100 each in staff time [19]. Wysa's participation in the **NHS Innovation Accelerator in 2023** further boosted its profile, connecting it with NHS commissioners nationwide [24]. **Lessons for OTOS:** even a highly innovative approach (AI chatbot) gained NHS adoption by rigorously addressing safety/data concerns and by quantifying its impact on service efficiency. Wysa also shows that being inexpensive and easily deployable can lead services to adopt it as a low-risk experiment initially, which then turns into recurring contracts when it proves its worth. OTOS, with its digital+human blended model, can similarly emphasize how it *extends the reach of clinicians* (e.g. one OTOS mentor can support many patients through the app, versus one-to-one traditional therapy) and thus helps services do more with finite resources. Like Wysa, OTOS should gather user feedback and outcome data in any pilot to show high acceptability and symptom improvement, persuading NHS partners that it's both popular with patients and effective.

Each of these examples – Sleepio, SilverCloud, Wysa – went from pilots to commissioned services by **demonstrating outcomes and cost-effectiveness, aligning with NHS standards, and leveraging support programs**. They confirm that the NHS is willing to fund digital therapeutics when presented with strong evidence and a clear fit to policy goals (whether it's improving access, saving money, or addressing unmet needs).

Actionable Guidance for OTOS

To maximize the chances of a successful NHS commissioning within 12–18 months, OTOS should take the following concrete steps:

1. **Design a Pilot That Yields Robust Evidence:** Plan the upcoming pilot with evaluation in mind. Define clear *clinical outcome measures* (e.g. reduction in relapse rates, improvements in ADHD symptom scores like ASRS, decreased anxiety/depression on GAD-7/PHQ-9, improved

functioning on EQ-5D or WSAS). Use validated scales and, if feasible, include a comparison group or historical control. This aligns with NICE's Evidence Standards Framework, ensuring you collect the level of data NHS decision-makers expect [2] [11]. Engage an academic partner or NIHR unit to oversee data collection and analysis for credibility. Aim to publish or at least openly report the results – transparency builds trust. By pilot's end, you should be able to **demonstrate clinically meaningful improvement** (e.g. "X% of OTOS users achieved sustained abstinence for 3 months, compared to Y% typically" or "ADHD inattentiveness scores improved by Z points"). Strong clinical outcomes are your foundation.

2. **Meet NHS Compliance Requirements (DTAC & Regulatory):** Prioritize hitting all the checklist items that NHS procurement will ask for. This means preparing your **DTAC evidence** – ensure OTOS has a CE/UKCA marking if needed (check if your software might be considered a medical device for managing a condition; if so, ensure regulatory compliance). Complete the DCB0129 clinical safety process: have a qualified Clinical Safety Officer produce a safety case identifying any risks in using OTOS (e.g. handling of crisis content) and how they're mitigated [9]. Implement rigorous data protection measures (ICO registration, NHS Data Security & Protection Toolkit self-assessment). Build interoperability using standards (for example, enable summary reports that could be shared with GPs or entered into patient records). Also, **accessibility** should be verified – perform user testing to ensure OTOS is usable for people with disabilities or low digital literacy. By pilot's end, aim to say: "OTOS has been assessed against the NHS Digital Technology Assessment Criteria and meets all required standards [7]." This pre-empts a major hurdle in procurement. Additionally, consider getting OTOS reviewed by ORCHA or listed in the NHS Apps Library – an endorsement here signals to NHS managers that OTOS is safe and effective. Essentially, **be "procurement-ready" on the compliance front**, so no one can object "we can't buy this because it hasn't passed our information governance."

3. **Quantify the Economic Benefit (Build the ROI Case):** Use pilot data to articulate a compelling cost-saving or cost-effective narrative. Track healthcare usage for participants (with appropriate consent/data sharing agreements in place). For example, monitor if participants use fewer crisis services or have shorter inpatient stays. If possible, collect baseline data from before OTOS intervention to compare post-intervention costs. Calculate the **cost per patient for delivering OTOS**, and then calculate **cost offsets**: "Using OTOS for 3 months costs £A, but saves £B in reduced service use, yielding £C net savings (or £D saved per £1 spent)." Even if OTOS doesn't immediately save cash, it might enable the system to handle more patients for the same cost (efficiency gain) – quantify that (e.g. "clinicians could manage 20% more patients with OTOS support"). Leverage examples like Sleepio – which saved ~£106 per patient by reducing other care [17] – or the **rehab comparison** – one month rehab vs 12 weeks of OTOS for 12 people – to illustrate potential scale of savings. You should also frame how OTOS can **prevent high-cost events**: for instance, if even 1 in 10 OTOS users avoids a £5,000 inpatient relapse treatment, that's an average £500 saving per user right there. Put these figures in an easy-to-digest format (graphs or one-page summaries) for commissioners. Consider using NICE's budget impact tool to project savings for an ICS population [21]. By the end of the pilot, create a draft **business case document** with all this data – something you can hand to an NHS commissioner showing the investment required and the returns. Speak the language of NHS finance: mention QALYs if appropriate (e.g. improved quality of life for patients has its own value), but especially **mention savings in the current financial year**, since NHS managers love in-year cost savings or at least cost-neutrality.

4. **Engage Early with Commissioners and Build Partnerships:** Don't wait until the pilot is over to talk about commissioning. Involve the local ICS (Cambridgeshire & Peterborough) stakeholders from the start – for example, invite a commissioner or ICS digital lead to sit on the pilot steering

group. This way they feel ownership and see interim results. Leverage any existing relationships: CPFT's medical director or the ICS mental health lead should be kept informed of progress. **Identify a champion** – perhaps a respected clinician (e.g. an addiction psychiatrist or an ADHD service lead involved in the pilot) – who can advocate for OTOS internally. Their testimonials about OTOS improving patients' engagement or outcomes will carry weight. In parallel, connect with the **Eastern AHSN** and the East of England NIHR network; they can provide advice and might showcase OTOS in regional innovation forums. As momentum builds, prepare a list of target ICSs for expansion (those with high ADHD and addiction prevalence). Start informal conversations with their commissioners *now*, using the pilot as a teaser ("We're doing this in Cambridge; if it works, we'd love to bring it to your patch next – what outcomes would you need to see?"). This not only readies your next market but also helps refine what data to emphasize. Essentially, treat commissioning as a co-development process with the NHS, not just a sales pitch at the end.

5. **Leverage Innovation Pathways (NIA, Accelerators, Funding Competitions):** Apply to structured programs that can fast-track OTOS. For example, once you have some pilot data, **apply for the NHS Innovation Accelerator** – being selected would give OTOS national visibility and mentoring on NHS scaling. The NIA has helped tools like Wysa gain contracts across multiple regions [24]. Similarly, look at the **DigitalHealth.London Accelerator** or regional equivalents (the next cycle may be open to companies outside London as well). These programs can hook you up with mentors who are NHS procurement experts or seasoned healthtech entrepreneurs. Monitor opportunities like the **NIHR i4i (Invention for Innovation) funding** or **SBRI Healthcare competitions** – these often have calls for mental health or addiction solutions. Securing a grant could fund a larger trial or even subsidize an extended pilot (for example, NIHR or SBRI might fund a Phase 2 where you implement OTOS in two ICSs to gather comparative data). Not only does this bring funding, but winning competitive awards builds credibility (NHS commissioners take note of SBRI winners or NIHR-backed projects as they've been peer-reviewed). Additionally, engage with initiatives like the **NHS Clinical Entrepreneur Programme** if any of your team are clinicians, or pitch at AHSN "Innovation Exchange" events. The more embedded you are in the NHS innovation ecosystem, the more likely you'll hear about commissioning opportunities and be seen as a serious, validated player.

6. **Focus the Value Proposition on Mental Health Outcomes:** When presenting OTOS, categorize it appropriately to fit NHS commissioning structures. Emphasize that OTOS addresses mental health and neurodevelopmental needs (the ADHD component) with an innovative approach to addiction recovery – a true *integrated care* model. This way, funding can potentially come from NHS mental health budgets that have remits for improving community care and reducing acute admissions. Many ICSs have a priority around "dual diagnosis" (co-occurring mental illness and substance misuse) because historically these patients fall through gaps. OTOS should pitch itself as **the solution for dual diagnosis patients** – improving engagement and outcomes where traditional services struggle. This aligns with NHS goals of holistic, person-centered care. In practical terms, this means when writing business cases or discussing with commissioners, frame outcomes in terms like "improved mental health stability," "reduced crisis contacts," and "supporting the NHS Long Term Plan objective of better community-based care for severe mental illness," rather than framing OTOS as a niche addiction app. If commissioners see it as part of the mental health toolkit, they can use ring-fenced mental health funding (which has been growing) to pay for it, instead of the cash-strapped public health addiction budgets. Also highlight how OTOS complements existing services: e.g. it can **work alongside NHS Adult ADHD clinics (improving titration adherence, providing psychoeducation)** and alongside substance misuse services (offering relapse prevention between counseling sessions). The integrated framing will make it attractive to ICS leads who are tasked with breaking silos between services.

7. **Plan for Scale and Practical Implementation:** Think ahead to what happens if an ICS says "yes, we'll fund it." Ensure you have answers for implementation questions: Who will refer patients to OTOS (or will it be self-referral)? How will OTOS integrate with existing care teams (e.g. will OTOS mentors join MDT meetings or share reports with clinicians)? What training or onboarding is needed for NHS staff? Have these operational details refined during the pilot so that post-commissioning roll-out is smooth and does not create extra work for the NHS staff (who may be change-averse). The easier you make it to deploy OTOS (technically and workflow-wise), the more likely the NHS will adopt it at scale. As part of procurement prep, also determine your pricing model for the NHS – many digital health companies use population-based pricing for ICS contracts (for example, £X per year for unlimited use within the ICS up to a certain population), which simplifies budgeting. Be ready to be flexible in contracting (e.g. pilot-to-full contract conversions, volume discounts if multiple ICSs join). Also, consider requirements like **Care Quality Commission (CQC)** registration if OTOS's mentor service constitutes a regulated activity (e.g. therapy) – if so, ensure you have or plan to obtain any necessary CQC status so that regulatory compliance is not a blocker.

By executing on these steps, OTOS can position itself as an evidence-based, cost-saving, and implementation-ready solution – exactly what NHS commissioners look for in innovations to invest in. The timeline is tight but achievable: within 12 months, complete the pilot and gather outcomes; by 18 months, use that data plus the partnerships forged to secure initial commissioning in the pilot ICS and ideally one or two additional regions (leveraging the early adopter effect). The path blazed by digital tools like Sleepio, SilverCloud, and Wysa shows that the NHS will embrace digital mental health innovations that **speak to its priorities (improved access, outcomes, and efficiency) and meet its standards**. With diligent evidence-gathering and stakeholder engagement, OTOS can transform from a promising pilot into a commissioned service delivering better outcomes for people with ADHD and addiction – while saving the NHS money and capacity – a true win-win outcome.

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